A.

C1 S1 C2 S2 C3 S3 C4 S4 C5 S5

spring cooling
summer drying

spring warming
summer wetting

spring cooling

spring warming

spring warming

Standardized climate level
(calibration SD from the modern mean)

1

0

-1

Spring temperature (CCA constraint) Winter temperature Summer precipitation (CCA constraint) Winter precipitation

B.

Fire-frequency ratio
(high fire > 1)

1.2
0.8
0.4

MORA

NOCA

10,000 8,000 6,000 4,000 2,000 0

cal BP